

An EXHAUST FAN That SWITCHES ON AUTOMATICALLY On Detecting Smoke

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Normally exhaust fans installed in kitchens, including restaurant kitchens, keep running continuously—whether smoke is present or not. Here is how you can build a small electronic device that switches on the exhaust fan automatically only when there

is smoke in the air. It switches the exhaust fan off when the smoke has been removed.

Circuit and working

Circuit diagram of the auto exhaust fan switch is shown in Fig. 1. It has 5V voltage regulator 7805 (IC1),

AVR microcontroller ATtiny85A (IC2), gas sensor MQ2 (SEN1), common-cathode bi-colour LED (BICO1), 5V single changeover relay (RL1), NPN transistor BC547 (T1), and a few other components.

The ATtiny85A turns the exhaust fan connected through the relay

on or off based on extent of smoke in the air. The resistance offered by MQ2 sensor changes with smoke concentration near the sensor and thus changes voltage at analogue input pin 3 (PB4) of IC1. Potentiometer VR1 is used to adjust the sensitivity of the sensor during calibration.

When the circuit is powered on, bi-colour LED BICO1 blinks in red colour for 20 seconds, indicating that the circuit is in the necessary pre-heat stage (see Fig. 1). After that the microcontroller continuously checks whether

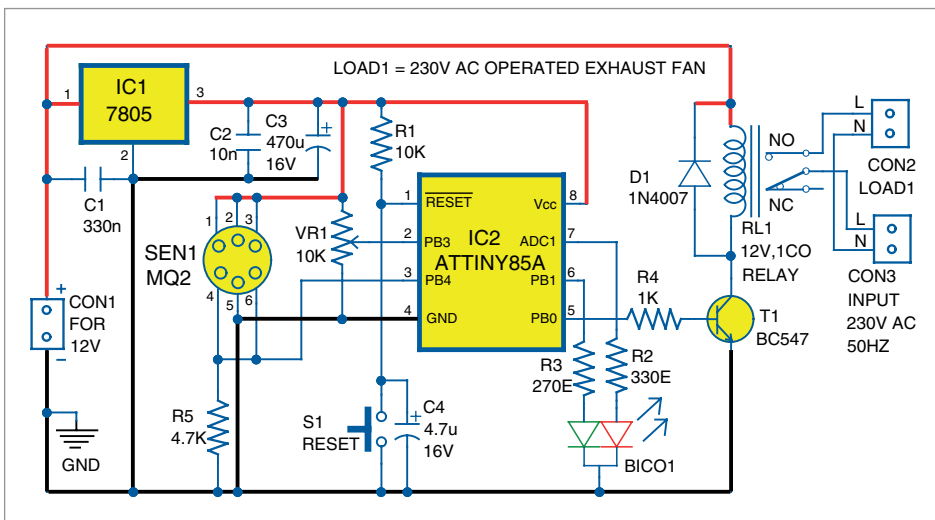


Fig. 1: Circuit diagram

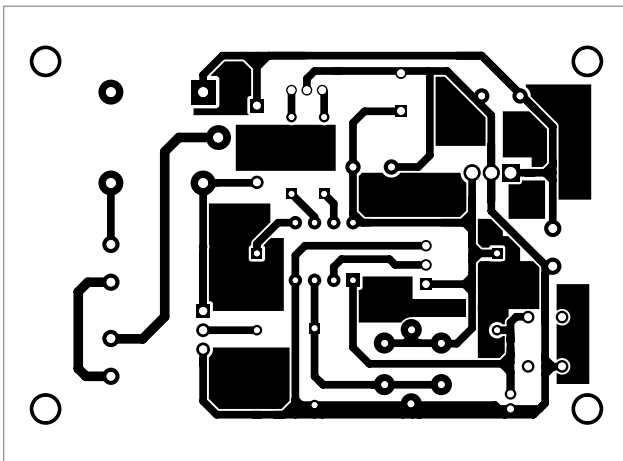


Fig. 2: Actual-size PCB layout

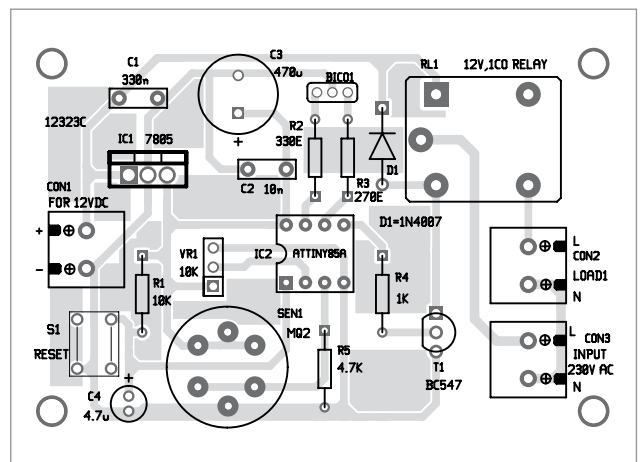


Fig. 3: Component layout of the PCB